

Relational Clocks and Causal Cartan Contraction through Arity Three in Pure-Weyl Gravity on a Berger Universe

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Abstract

We study whether cylinder-time translation can remain gauge after a healthy matter clock and the first nonlinear vertices are included in pure-Weyl gravity. On $\mathbb{R} \times S^3$ with a Berger spatial metric we exhibit an open family of exact, smooth, non-conformally-flat solutions sourced by two standard-sign conformal scalars. Their rotating phase is an everywhere timelike clock, the quartic potential is positive, and the internal clock momentum is nonzero. Nevertheless, at fixed couplings the linearized lapse constraint and compact spatial averaging force its variation to vanish on every smooth solution tangent. The covariant identity $\Omega_{\text{total}}(\delta, \mathcal{L}_D) = \omega \delta Q_R$ therefore makes the helical generator D a presymplectic degeneracy on this declared phase space.

We then formulate the same conclusion homologically. The complete gauge-fixed classical BV presentation has 54 rows and a support-local cyclic retract onto a 26-row retained complex. Advanced and retarded chain contractions lift to all rows. Differentiating the covariant master action gives complete arbitrary-input four-dimensional operations q_2 and q_3 , with the required L_∞ identities, cyclicity and local D -equivariance. The unary and binary Cartan primitives extend to a cyclic ternary primitive: the arity-three source closes by the graded Jacobi identity, and a causal raw primitive becomes cyclic under the exact C_4 Reynolds projector. Thus the D action is causally null-homotopic through arity three.

The result is classical and scoped to one compact fixed-coupling Berger sector. It is not an all-orders nonlinear theorem, a Hadamard construction, a quantum-master-equation result, an anomaly calculation, or a claim that a nonzero clock momentum is a freely variable boundary charge.

Authorship and computational inputs

The named model author produced the mathematical programme, derivations, verification code, referee responses, and manuscript. Asger Alstrup Palm commissioned and directed the work, initiated the questions, and served as non-technical orchestrator and corresponding human contact; he claims no technical contribution. Repository scripts generate the certificates and sparse coefficient tables used as computational inputs. They are not additional authors. This article has no generated TeX input.

*OpenAI model. Contributed the research programme, mathematical direction, derivations, machine verification, referee analyses, and manuscript. The project was commissioned and directed by Asger Alstrup Palm (asger@area9.dk), who initiated the questions, served as the non-technical orchestrator and corresponding human contact, but claims no technical contribution.

Contents

1	Question, phase space and theorem boundary	2
2	A positive clock on an exact Berger background	3
3	Nonzero clock momentum and vanishing tangent charge	4
4	Gauge explanation and relational evolution	5
5	The complete causal BV complex	5
6	Action-derived interaction tensors	6
7	Causal Cartan contraction through arity three	6
7.1	Support statement	7
8	Interpretation, failures and open directions	8
9	Reproducibility and claim map	8

1 Question, phase space and theorem boundary

There are two superficially conflicting requirements for a relational clock. Its phase must evolve and possess a nonzero conjugate momentum, yet the total Hamiltonian generator should remain a constraint on a closed universe. A matter phase with zero momentum is not a clock, while a freely variable nonzero Hamiltonian charge would make time translation a physical symmetry rather than proper gauge. The distinction is made by the pullback of the charge variation to the allowed solution space, not by the background value of the charge. We use the BV description of gauge systems [2, 3], the covariant phase-space pairing [4], and causal chain contractions in the sense of Green-hyperbolic complexes [6].

We work on

$$M = \mathbb{R} \times S^3 \tag{1}$$

with no spatial boundary. Couplings are fixed. The phase space used for the charge theorem is the space of smooth linearized solutions about the exact Berger clock background, modulo the declared local diffeomorphism and Weyl gauge transformations. The nonlinear statement is a formal Taylor statement about the classical BV vector field through arity three at that same background. These are related statements, but they are not identical.

Theorem 1.1 (Scoped Berger clock and Cartan theorem). *Fix $\alpha_B > 0$ and a point of the positive Berger branch described in Section 2. Then:*

1. *the background carries a smooth standard-sign conformal-scalar phase clock with positive quartic potential, timelike phase gradient and nonzero internal momentum;*
2. *on the smooth fixed-coupling linearized covariant phase space,*

$$\Omega_{\text{total}}(\delta, \mathcal{L}_D) = 0 \tag{2}$$

for every allowed tangent δ ;

3. the complete 54-row gauge-fixed classical BV complex admits advanced and retarded chain contractions with causal support;
4. if q_1, q_2, q_3 are the first three action-derived Taylor operations, then there are cyclic causal Cartan components $\iota_D^{(1)}, \iota_D^{(2)}, \iota_D^{(3)}$ satisfying the Cartan equation through arity three.

Consequently D is proper gauge at the linearized covariant level and is causally null-homotopic through the first ternary BV interaction test.

The four assertions are certified respectively by claims P09-C1–P09-C10 in the companion computational supplement. The theorem does not assert that the Cartan contraction extends to all Taylor orders or to a renormalized quantum theory.

2 A positive clock on an exact Berger background

Let σ_i be a left-invariant coframe on S^3 and set

$$g = -dt^2 + a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2, \quad q = \frac{c^2}{a^2}. \quad (3)$$

The matter fields are two real conformal scalars with positive target metric, written in polar form as

$$T_1 = \rho \cos(\omega t), \quad T_2 = \rho \sin(\omega t). \quad (4)$$

Their phase $\theta = \omega t$ obeys $g^{-1}(d\theta, d\theta) = -\omega^2 < 0$ and is therefore an everywhere timelike local clock. The temporal-diffeomorphism/Weyl incidence determinant is $\rho^2\omega$, so the clock slice is regular on the branch below.

Our action convention is

$$S = \int_M \sqrt{-g} \left[\frac{\alpha_B}{8} C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} - \frac{1}{2} \sum_{A=1}^2 \nabla_\mu T_A \nabla^\mu T_A - \frac{R}{12} \rho^2 - \frac{\lambda}{4} \rho^4 \right]. \quad (5)$$

With $T_{\mu\nu} = -2\delta S_m / \delta g^{\mu\nu}$, its metric equation is

$$\alpha_B B_{\mu\nu} = T_{\mu\nu}. \quad (6)$$

Proposition 2.1 (Exact positive Berger family; P09-C1, P09-C2). *For*

$$\frac{5 - \sqrt{21}}{2} < q < \frac{1}{4} \quad (7)$$

there is an exact solution of the scalar and metric equations with

$$\rho^2 = \frac{2\alpha_B(1-4q)}{a^2}, \quad \omega^2 = \frac{q}{4a^2(1-4q)}, \quad \lambda = -\frac{q^2 - 5q + 1}{6\alpha_B(1-4q)^2} > 0. \quad (8)$$

The scalar kinetic metric is positive, the quartic potential is bounded below, and the energy density dominates both principal pressures.

Proof. In the orthonormal Berger frame the independent Bach components are

$$B_{00} = \frac{(a^2 - c^2)^2}{6a^8},$$

$$B_{11} = B_{22} = \frac{(a^2 - c^2)(a^2 - 3c^2)}{6a^8}, \quad (9)$$

$$B_{33} = \frac{(a^2 - c^2)(5c^2 - a^2)}{6a^8}. \quad (10)$$

Substitution of (4) into the scalar equation and the three independent components of (6) gives (8). The interval (7) makes ρ^2 , ω^2 and λ positive. Direct substitution gives

$$\varepsilon = \frac{\rho^2(1-q)^2}{12a^2(1-4q)},$$

$$p_h = \frac{\rho^2(1-q)(1-3q)}{12a^2(1-4q)}, \quad p_v = \frac{\rho^2(1-q)(5q-1)}{12a^2(1-4q)}, \quad (11)$$

from which positivity and $\varepsilon \geq p_h, p_v$ follow on the stated interval. The coefficientwise verification also checks the lapse, horizontal-scale and vertical-scale variations of the Weyl-square action. $\square$

The rational point

$$a = 1, \quad q = \frac{9}{40}, \quad \alpha_B = 5, \quad \rho = 1, \quad \omega = \frac{3}{4}, \quad \lambda = \frac{119}{480} \quad (12)$$

is used for the exact all-row BV computations. It is a representative of the open branch, not a finite-mode replacement for the four-dimensional operations.

3 Nonzero clock momentum and vanishing tangent charge

The matter action has an internal $O(2)$ rotation

$$R(T_1, T_2) = (-T_2, T_1) \quad (13)$$

with current

$$j_R^\mu = T_1 \nabla^\mu T_2 - T_2 \nabla^\mu T_1. \quad (14)$$

For the Maurer–Cartan volume convention $\int_{S^3} \sigma_1 \wedge \sigma_2 \wedge \sigma_3 = 16\pi^2$, the background charge is

$$Q_R = 16\pi^2 \alpha_B q \sqrt{1-4q} > 0. \quad (15)$$

This is the canonical momentum conjugate to the phase θ . Thus the clock evolves and carries a genuine momentum.

The helical generator D acts trivially on the stationary metric and rotates the scalar pair:

$$\mathcal{L}_D g = 0, \quad \mathcal{L}_D T = \omega R T. \quad (16)$$

The action-derived covariant presymplectic form then gives the exact identity

$$\Omega_{\text{total}}(\delta, \mathcal{L}_D) = \omega \delta Q_R. \quad (17)$$

The pure-Weyl cross term vanishes because $\mathcal{L}_D g = 0$; the matter term is the variation of the Noether charge (15). This establishes P09–C3, but by itself it does not decide whether D is gauge. The decision depends on the allowed tangent space.

Proposition 3.1 (Fixed-coupling charge theorem; P09-C4). *On the smooth fixed- α_B , fixed- λ linearized solution space about the positive Berger clock background,*

$$\delta Q_R = 0. \quad (18)$$

Hence (2) holds and D is a proper-gauge direction on this phase space.

Proof. Retain the lapse before varying the homogeneous reduced action and fix only the common Weyl scale. The exact linearized lapse equation on the branch is

$$\delta E_N = -\frac{\alpha_B q^{3/2}}{2} \frac{\delta Q_R}{Q_R}. \quad (19)$$

Its coefficient is nonzero throughout (7), so every homogeneous allowed tangent has $\delta Q_R = 0$.

The compact spatial isometry group preserves the background and the linearized equations, while δQ_R is an invariant linear functional. If a smooth inhomogeneous solution tangent had nonzero δQ_R , averaging it over the compact group would produce a homogeneous solution tangent with the same nonzero value, contradicting (19). Thus (18) holds on the complete declared smooth tangent space. Equation (17) gives the result. $\square$

The compact averaging step is the matter-clock analogue of the familiar linearization-stability constraints on compact Cauchy surfaces [5]; here the exact lapse identity (19) supplies the decisive coefficient.

Remark 3.2 (What is fixed). The interval (7) parametrizes theories through the product $\alpha_B \lambda$; it is not a modulus inside one fixed theory. At fixed couplings the stationary branch does not furnish a freely variable q or Q_R . There is therefore no contradiction between $Q_R > 0$ and $\delta Q_R = 0$ on allowed tangents.

4 Gauge explanation and relational evolution

The phase θ changes along D , but a gauge transformation of the total system need not leave every partial observable fixed. Relational quantities compare another field to the clock. If X is a scalar quantity with $DX \neq 0$, then locally one may evaluate X on the slice $\theta = \theta_0$ or form combinations invariant under the simultaneous shift of X and θ . The background momentum (15) makes this clock slice nondegenerate, while Proposition 3.1 says that the total helical motion is a gauge orbit of the compact constrained system.

This result is sector dependent. A boundary Hamiltonian, nonzero flux, or a phase space allowing $\delta Q_R \neq 0$ would change the classification of D . Nothing here promotes the compact Berger verdict to asymptotically flat, de Sitter, anti-de Sitter, or bounded regions.

5 The complete causal BV complex

Let $(\mathcal{C}_{54}, q_1)$ denote the gauge-fixed classical BV complex at the rational point (12). Its row degrees have multiplicities

$$5 \text{ in degree } -1, \quad 22 \text{ in degree } 0, \quad 22 \text{ in degree } 1, \quad 5 \text{ in degree } 2. \quad (20)$$

The rows include the metric and clock variables, diffeomorphism and Weyl ghosts, antifields, antighosts and multipliers. A gauge-fermion canonical shear preserves the frozen odd Darboux pairing and gives a support-local cyclic contraction

$$\pi_{\text{cl}} \iota_{\text{cl}} = \text{id}, \quad \text{id} - \iota_{\text{cl}} \pi_{\text{cl}} = q_1 S_{\text{cl}} + S_{\text{cl}} q_1 \quad (21)$$

onto a retained 26-row complex. This is claim P09-C5.

The retained complex has advanced and retarded chain contractions $\Lambda_{26,\pm}$. Lifting them through (21) gives

$$\Lambda_{54,\pm} = S_{\text{cl}} + \iota_{\text{cl}}\Lambda_{26,\pm}\pi_{\text{cl}}, \quad (22)$$

and therefore

$$q_1\Lambda_{54,\pm} + \Lambda_{54,\pm}q_1 = \text{id}_{54}, \quad \text{supp}(\Lambda_{54,\pm}f) \subseteq J^\pm(\text{supp } f). \quad (23)$$

All maps in the retract are differential or pointwise algebraic. No inverse Laplacian, inverse curl, transverse-traceless projector, or mode projector is used. The complementary-degree adjoint relation and D -equivariance also lift. These statements comprise P09-C6.

6 Action-derived interaction tensors

Expand the classical BV vector field in the suspended graded-symmetric factorial convention,

$$Q = q_1 + q_2 + q_3 + O(\Phi^4). \quad (24)$$

The operations q_2 and q_3 are obtained by differentiating the covariant Weyl-clock master action on arbitrary four-dimensional jets and transporting the result through the same clock and gauge-fermion canonical maps used at arity one. They are not fits to residual modes or a harmonic cutoff.

The arity-two and arity-three parts of $Q^2 = 0$ are

$$[q_1, q_2] = 0, \quad (25)$$

$$[q_1, q_3] + \frac{1}{2}[q_2, q_2] = 0. \quad (26)$$

The common action origin supplies cyclicity with respect to the odd Darboux pairing. The stationary invariant-frame generator $D = e_0$ acts linearly, so its higher Taylor components vanish through this order,

$$L_D^{(2)} = L_D^{(3)} = 0, \quad (27)$$

and exact coefficientwise verification gives

$$[D, q_2] = [D, q_3] = 0. \quad (28)$$

Claims P09-C7 and P09-C8 bind these statements to the portable all-row coefficient exports.

7 Causal Cartan contraction through arity three

Write

$$\iota_D = \iota_D^{(1)} + \iota_D^{(2)} + \iota_D^{(3)} + \cdots \quad (29)$$

for a degree-(-1) Cartan homotopy. At unary order,

$$[q_1, \iota_D^{(1)}] = D. \quad (30)$$

A one-sided raw solution follows from (23); averaging it with its convention-correct cyclic adjoint yields a cyclic unary primitive with support in the two-sided causal hull.

At binary order define

$$A_D^{(2)} = [q_2, \iota_D^{(1)}]. \quad (31)$$

Equations (25) and (28) make $A_D^{(2)}$ q_1 -closed. A raw causal primitive is cyclicized with $\text{Cyc}_3 = (1 + \tau + \tau^2)/3$, giving

$$[q_1, \iota_D^{(2)}] = -[q_2, \iota_D^{(1)}]. \quad (32)$$

This is P09-C9.

The ternary source is

$$A_D^{(3)} = [q_3, \iota_D^{(1)}] + [q_2, \iota_D^{(2)}] - L_D^{(3)}. \quad (33)$$

Lemma 7.1 (Closure of the ternary source). *The complete arbitrary-input source (33) obeys*

$$[q_1, A_D^{(3)}] = 0. \quad (34)$$

Proof. Let $\delta = [q_1, \cdot]$. Using (26), (30) and (32), the only nonzero formal channels in $\delta A_D^{(3)}$ are

$$-\frac{1}{2}[[q_2, q_2], \iota_D^{(1)}] + [q_2, [q_2, \iota_D^{(1)}]] - [q_3, D]. \quad (35)$$

Graded Jacobi gives

$$[q_2, [q_2, \iota_D^{(1)}]] = \frac{1}{2}[[q_2, q_2], \iota_D^{(1)}], \quad (36)$$

and $[D, q_3] = 0$ annihilates the last term. Thus the normalized coefficients are $-1/2 + 1/2 = 0$ and 0 in the two independent channels. $\square$

Because cyclic coderivations are closed under their graded bracket, $A_D^{(3)}$ is cyclic. The causal chain contraction supplies a raw primitive

$$R^{(3)} = -\Lambda_{54,+} A_D^{(3)}, \quad \delta R^{(3)} = -A_D^{(3)}. \quad (37)$$

Pair the output with the odd Darboux form before cyclicizing. The resulting four-linear tensor carries the standard Koszul C_4 action and the exact Reynolds projector

$$\text{Cyc}_4 = \frac{1}{4}(1 + \tau + \tau^2 + \tau^3). \quad (38)$$

Since $\tau^4 = 1$, $\text{Cyc}_4^2 = \text{Cyc}_4$. Cyclicity of q_1 makes Cyc_4 commute with δ , while cyclicity of the source gives $\text{Cyc}_4 A_D^{(3)} = A_D^{(3)}$. Therefore

$$\iota_{D,\text{cyc}}^{(3)} = \text{Cyc}_4 R^{(3)}, \quad [q_1, \iota_{D,\text{cyc}}^{(3)}] = -A_D^{(3)}. \quad (39)$$

This proves the ternary part of Theorem 1.1 and P09-C10.

The tensorized C_4 convention matters. The actual 54-row pairing has 27 negatively oriented dual slots. Its duality involution, degree-one pairing and odd skew orientation are checked on every row. The C_4 group law is exhausted over all actual degree classes, covering 978,736 admissible degree-zero row quartets with no defect.

7.1 Support statement

The Taylor operations and pairing are support-local. Green contractions and their cyclic adjoints are causal. Consequently

$$\text{supp}(\iota_{D,\text{cyc}}^{(3)}(f, g, h)) \subseteq J(\text{supp } f \cup \text{supp } g \cup \text{supp } h). \quad (40)$$

The cyclic primitive is not asserted to be separately retarded or advanced. The one-sided chain contractions (23) retain that stronger property; cyclic averaging generally does not.

8 Interpretation, failures and open directions

The result resolves one narrow version of the clock question. A healthy matter clock can have $Q_R > 0$ while the total helical generator remains gauge, because the constraint fixes Q_R within one theory and annihilates its pullback differential. The all-row BV calculation shows that this is not only a homogeneous minisuperspace coincidence: the linearized covariant phase space, causal complex and first interaction recurrences give compatible answers.

The following conclusions are not drawn.

1. No all-orders nonlinear Cartan contraction is claimed. Arity four and higher remain separate gates if the complete master action requires them.
2. No Hadamard two-point function, microlocal spectrum condition, renormalized composite, anomaly cancellation or quantum master equation is constructed.
3. No positive graviton Hilbert space or interacting probability theorem follows from the positive clock matter.
4. No noncompact or bounded phase space is covered. There D may acquire a boundary charge or flux and cease to be proper gauge.
5. Relational-redshift fixtures are applications, not part of the main theorem unless their independent propagation and backreaction gates close.

The correct continuation is therefore not to advertise an unrestricted solution of the problem of time. It is to test whether the same Cartan structure survives higher Taylor order, renormalized Ward identities and phase spaces with physical boundaries.

9 Reproducibility and claim map

The companion file `09-relational-clocks-berger-d-cartan-computational-supplement.tex` contains the focused claim ledger, exact certificate paths, coefficient inventory, replay commands and freeze checklist. Its machine-readable source is `../d_quotient_classical/certificates/PAPER_09_BERGER_CLAIM_TABLE.json`. The verifier checks strict Draft 2020–12 schema validity, certificate hashes, required true and false flags, and occurrence of every claim identifier in the manuscript sources.

The expensive q_3 action expansion is not silently rerun by the paper build. Its content-addressed Tier-2 receipt records the derivation, canonical transport, row-bounded identity replay and independent portable consumer. Fast paper checks import that frozen artifact by hash. A paper release may be marked `THEOREM_FROZEN` only after clean-tree replay and nonlinear-team sign-off; this working draft is not yet so marked.

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